

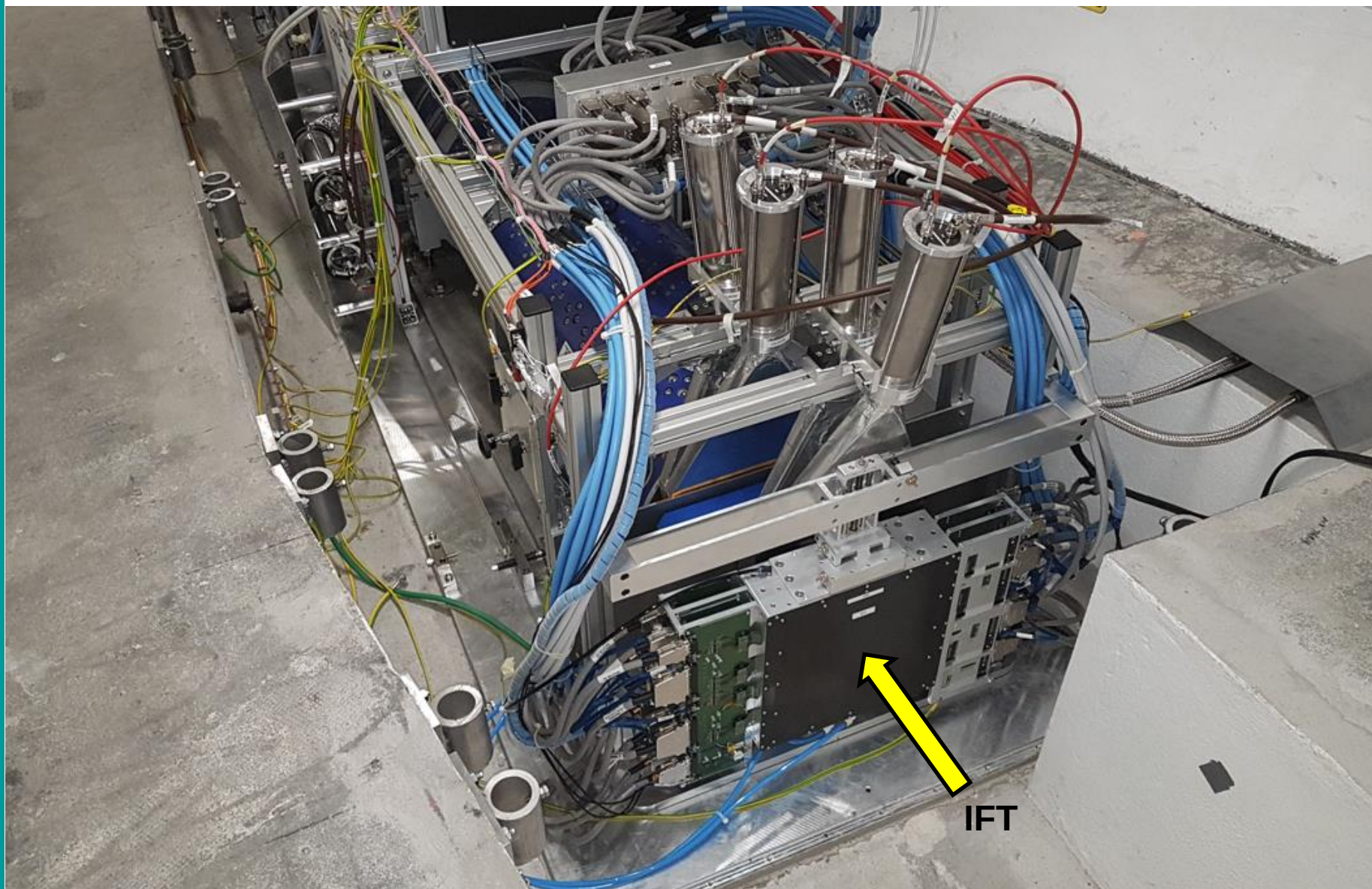


UNIVERSITÉ
DE GENÈVE

FACULTÉ DES SCIENCES
Département de physique
nucléaire et corpusculaire



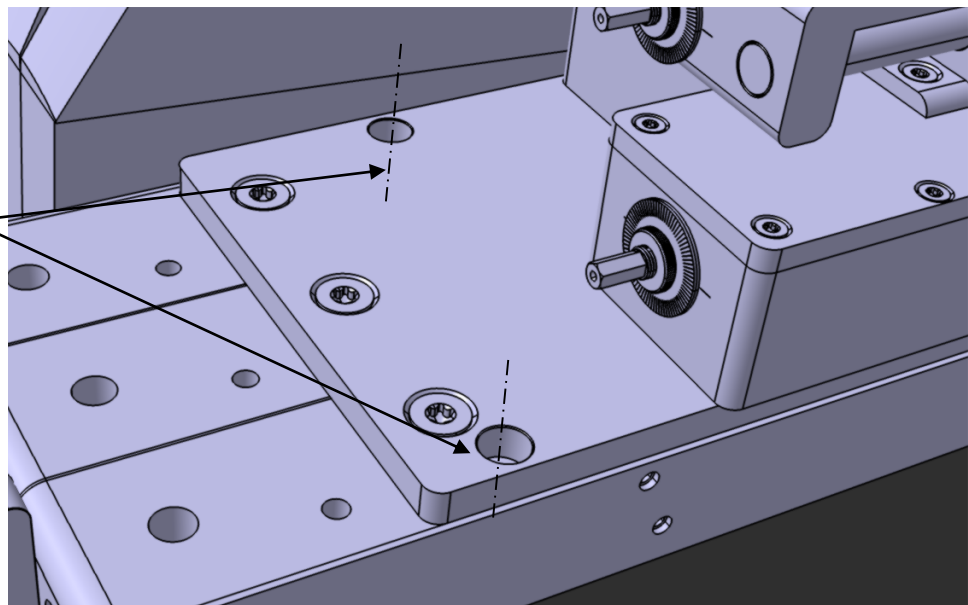
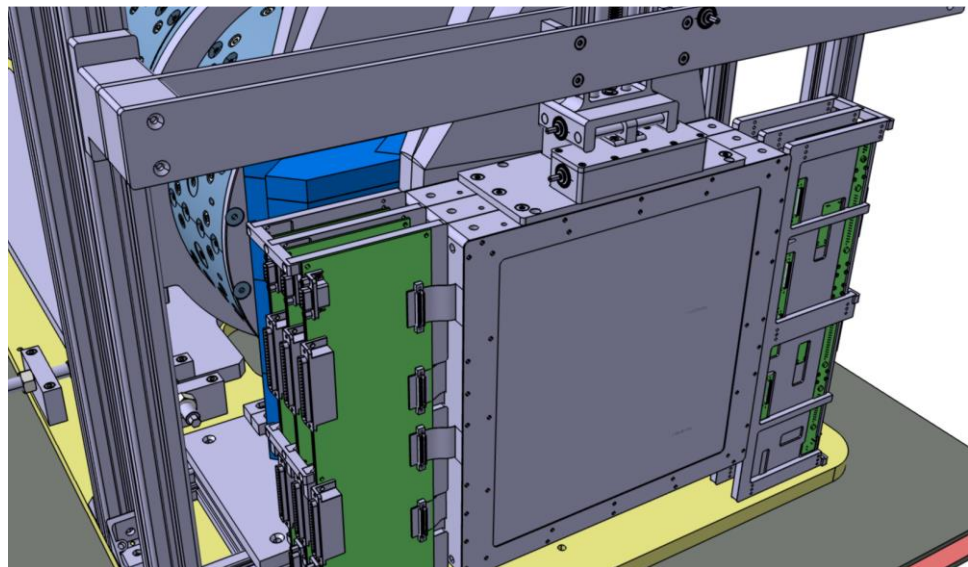
IFT Survey & Metrology



Frank Cadoux, FASER IFT, the 10th of December 2021

IFT Survey & Metrology

- Surveyed by CERN on Nov. 25
- IFT pre aligned, but with cables connected
- Not easy since cables blocked the access at the bottom side (stabilization system, see next)
- Adding some stress to the whole mechanics
- In ideal case, Survey & tuning would have been easier BEFORE cabling



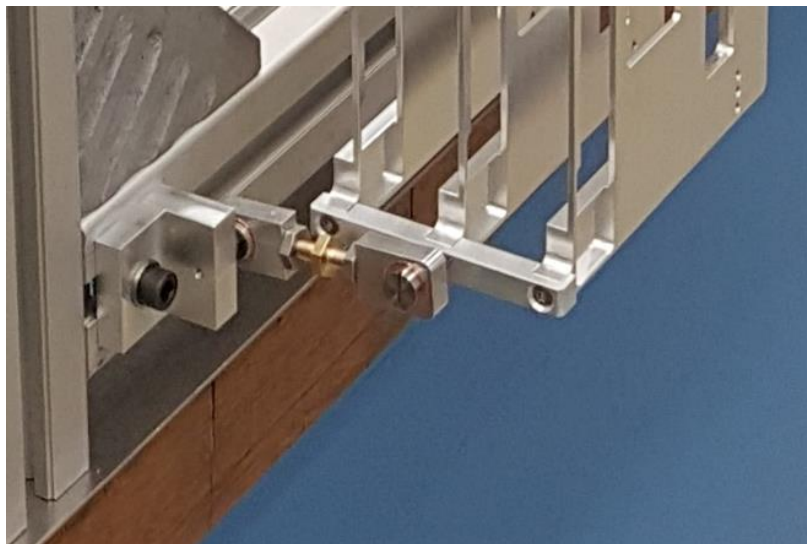
Target location



IFT Survey & Metrology

- Rehearsal @UNIGE on a test bench
- Stabilization mechanism
- On IFT structure, bottom side (hanging by the top part, for tuning)
- 2 fixed corners (tunable...)
- Due to lack of space, compact system...
- Limited number of degrees of freedom
- 2 translations, 3 rotations.
- Access issue with cables....

Corner fixation (rigid)



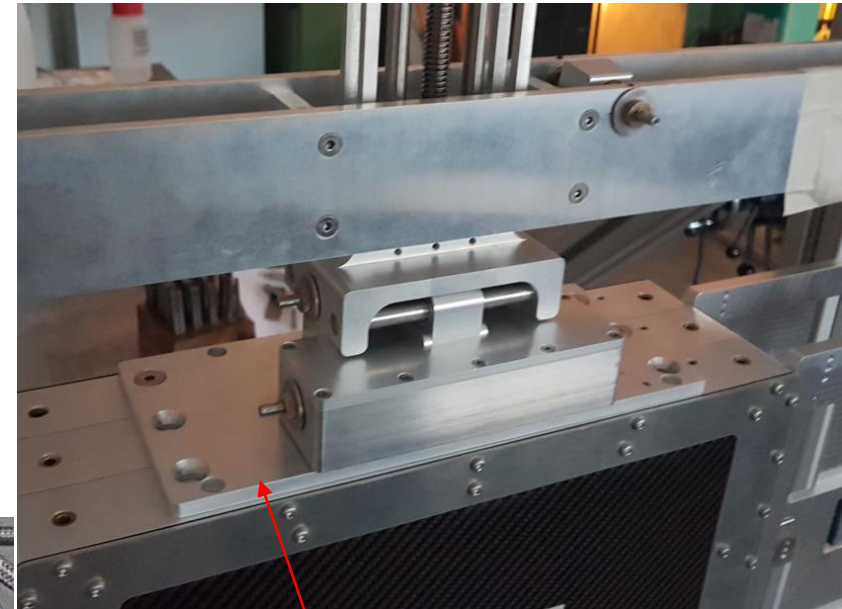
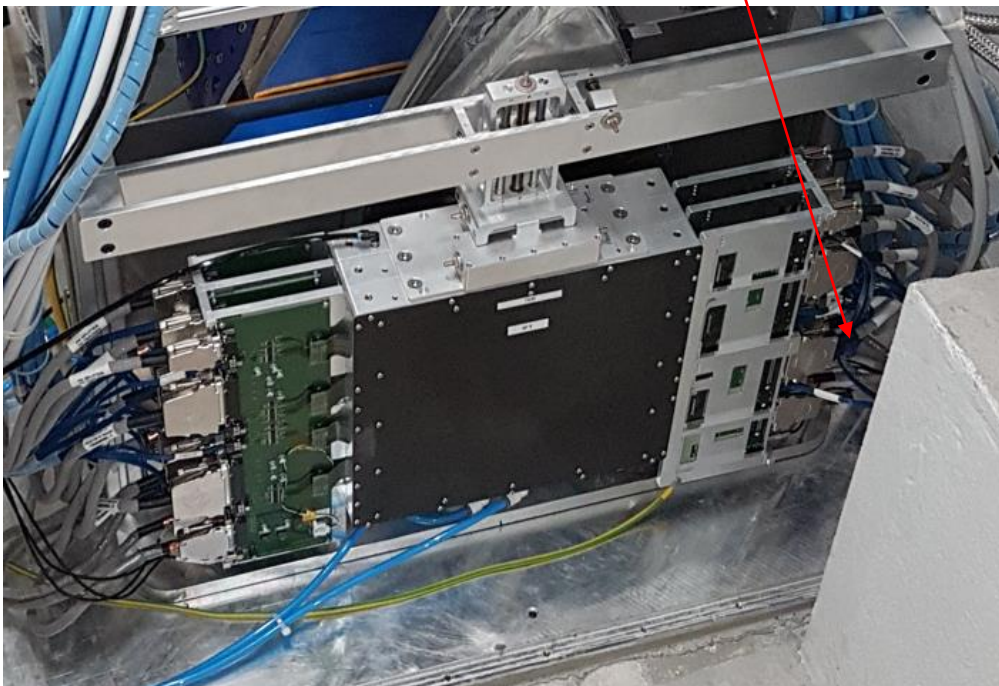
Tuning system



IFT Survey & Metrology

- Multi axes tuning system, attached to Upper Frame
- Fine movements, smooth...
- BUT too high load due to cables
- Partially de cabled (bottom row)

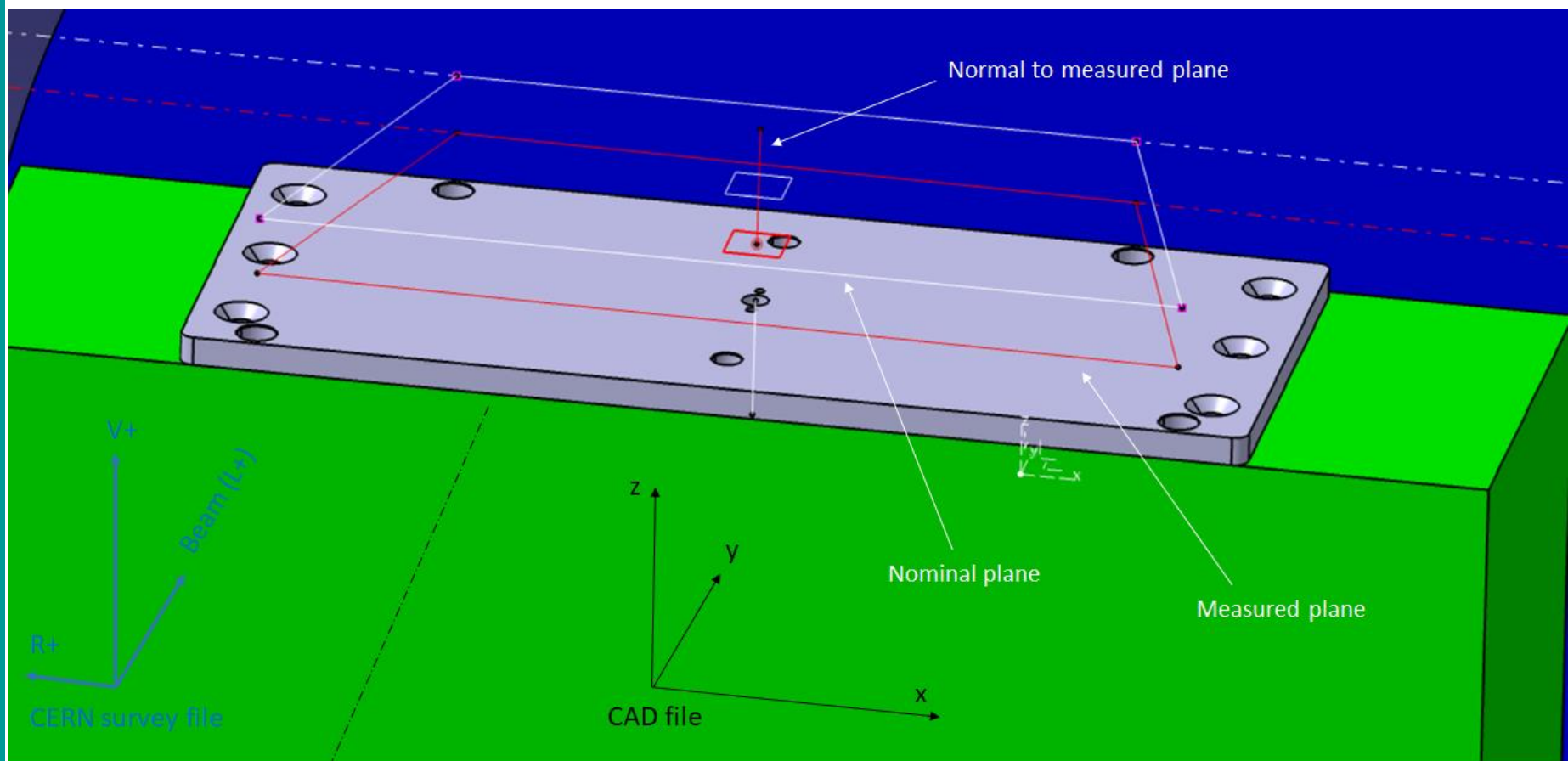
Painful access



Main I/F plate
(metrology vs Survey)

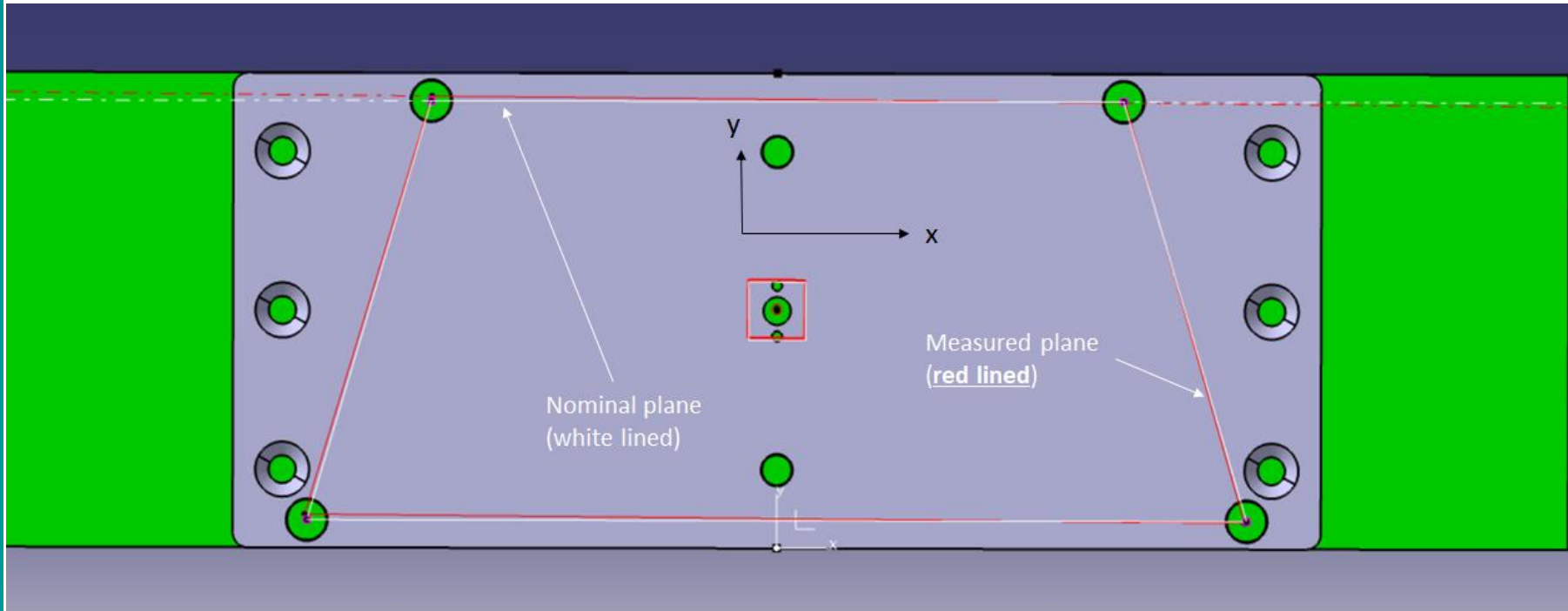
IFT Survey & Metrology

- Surveyed on 4 positions onto the I/F plane
- CAD is in its nominal position (**no vertical shift of 12mm** as decided for 3 TRK Stations!)
- Green volume is the simplified IFT



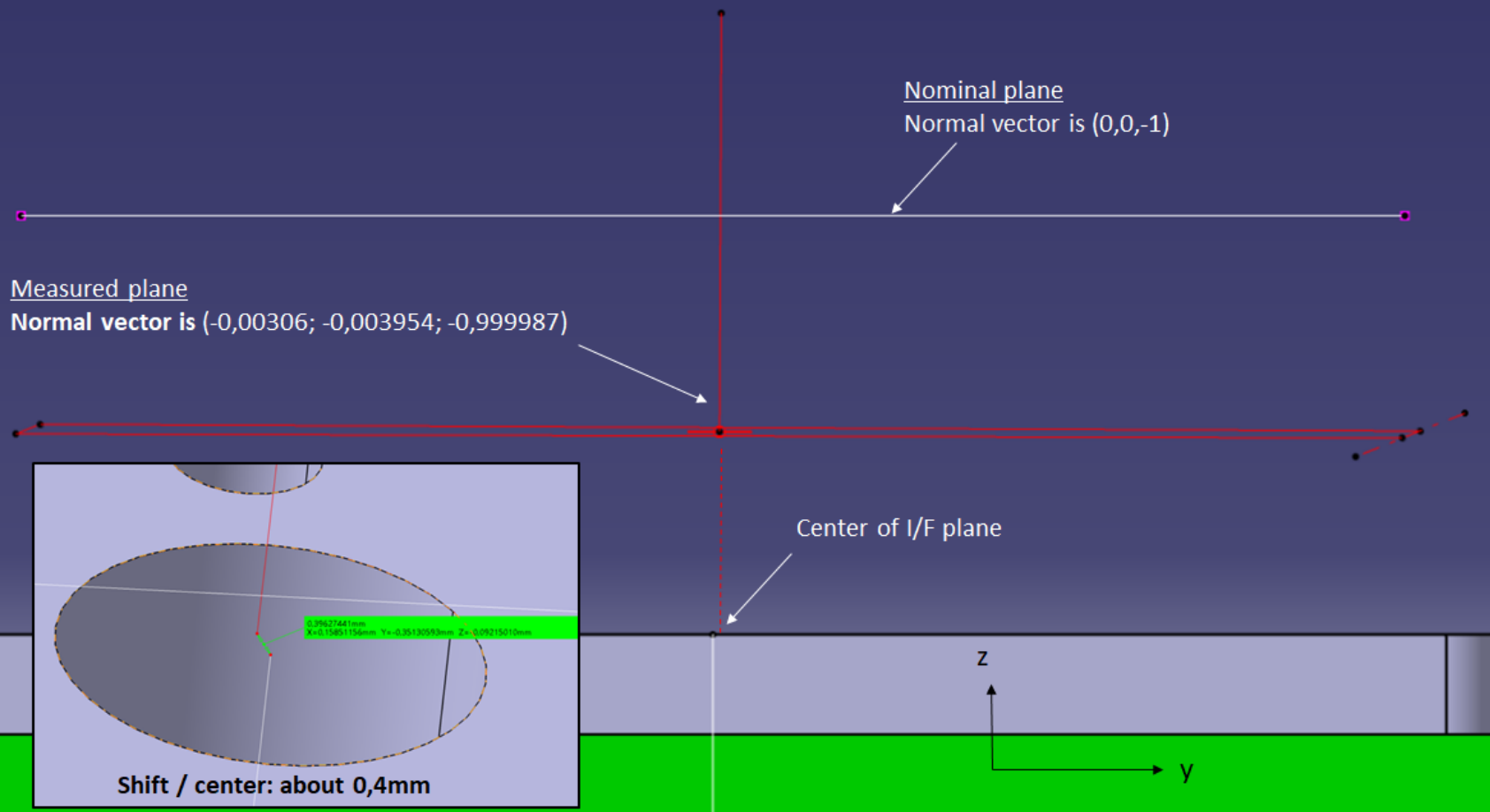
IFT Survey & Metrology

- Survey seen from the TOP
- Small rotation / z axis (difference between white and red)



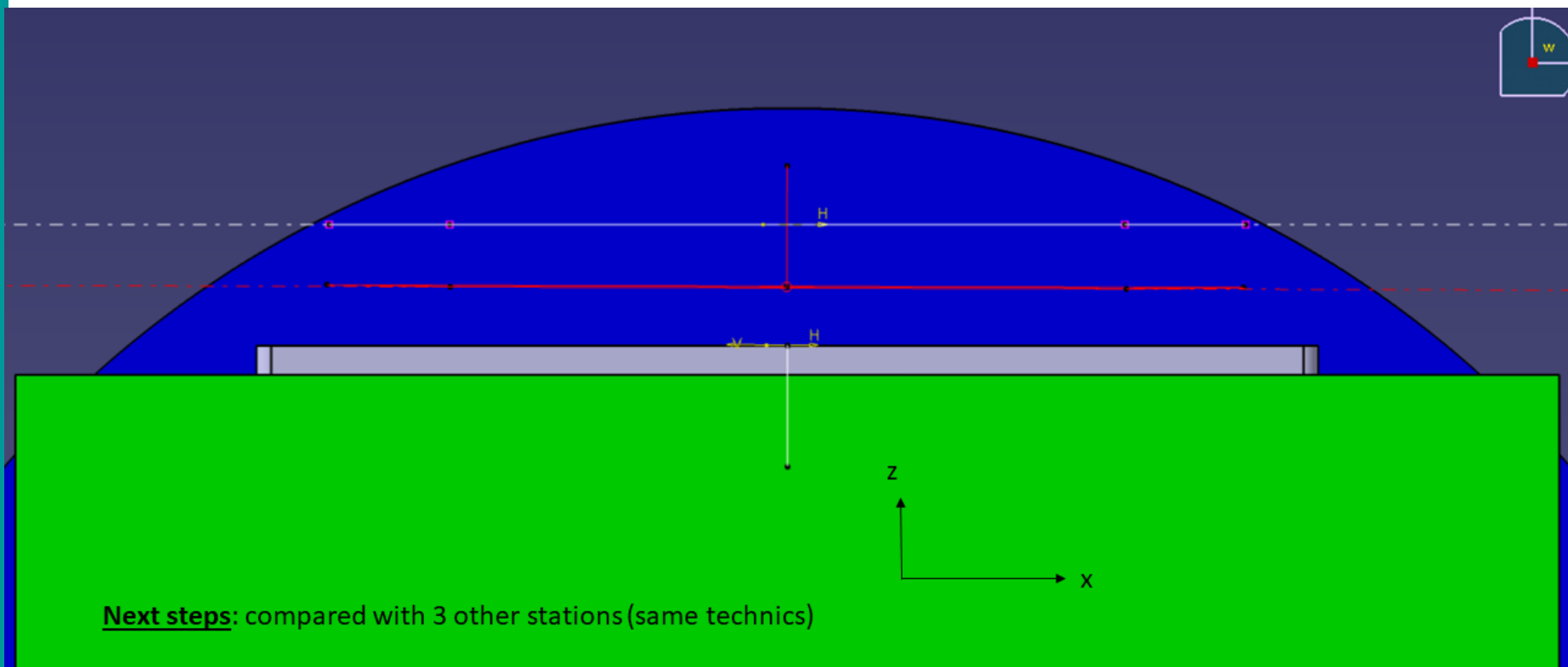
IFT Survey & Metrology

- Survey seen from the side
- Small rotation / y, z axes (difference between white and red)



IFT Survey & Metrology

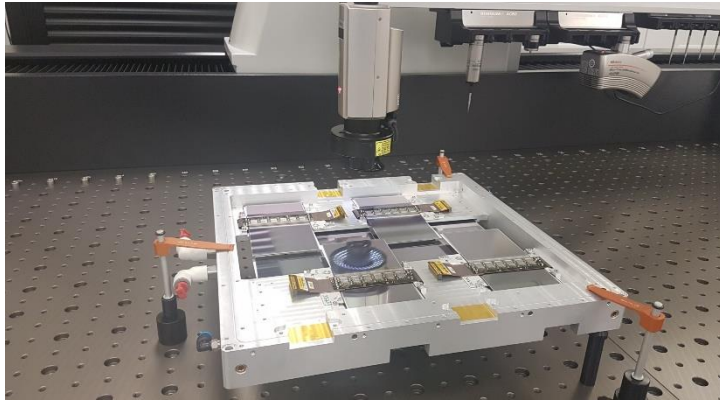
- Survey seen from the beam axis (LoS axis)
- Hard to see the rotations here... only the main vertical shift





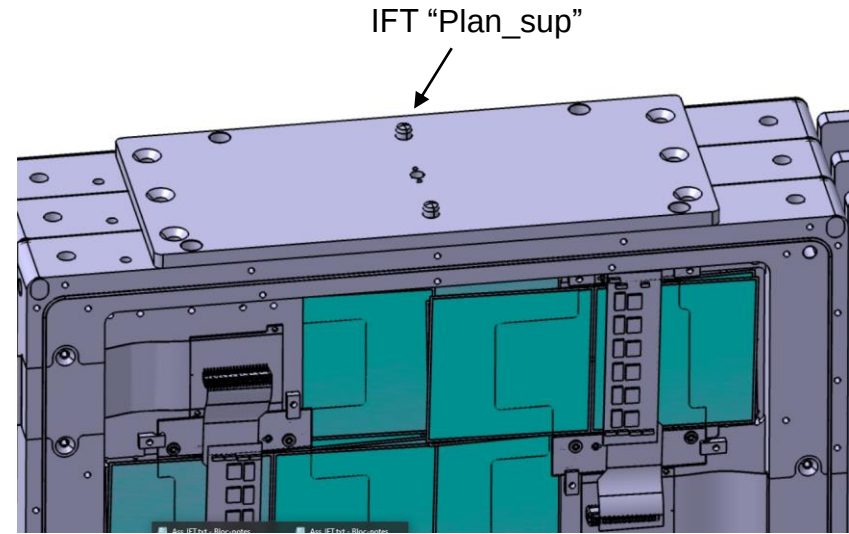
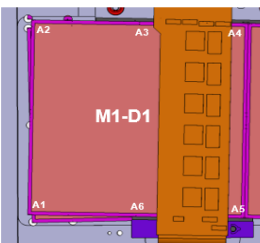
IFT Survey & Metrology

- IFT Metrology @UNIGE in 2 steps
 - Plane by plane (#1)
 - 3 planes assembled with top plate (#2)



Metrology#1: IFT Plane

Tol.PosNoTol.ElemNaTol.ElemNTol.TolNameTol.NomiTol.LowerTolTol.UpperTolTol.Tol.Deviation						
Plan_ref_F1000nt	Plan, it,	0.000	0.000	0.100	0.067	0.067
Plan_ref_A1002k	Plan, it,	0.000	0.000	0.100	0.032	0.032
Plan_ref_G1003emod1	Plan, it,	0.000	0.000	0.100	0.045	0.045
Plan_ref_G1004emod2	Plan, it,	0.000	0.000	0.100	0.063	0.063
Plan_ref_11001erfac	Plan, it,	0.000	0.000	0.100	0.012	0.012
Plan_ref_11001erfacPerpendicu	0.000,	0.000	0.000	0.100	0.056	0.056
Plan_ref_11001erfacPerpendicu	0.000,	0.000	0.000	0.100	0.061	0.061
Plan_ref_A1002k	Parallel, lis	0.000	0.000	0.100	0.030	0.030
Plan_ref_11001erfacAngle X	0:00:00	-0:30:00	0:30:00	0:00:38	0:00:38	0:00:38
Plan_ref_11001erfacAngle Y	0:00:00	-0:30:00	0:30:00	90:00:01	90:00:01	90:00:01
Plan_ref_11001erfacAngle Z	0:00:00	-0:30:00	0:30:00	89:58:22	89:59:22	89:59:22
Plan_ref_11001erfacDistance	160.000in	-0.100	0.100	160.022	0.022	0.022
Plan_ref_A1002k	Angle X	0:00:00	-0:30:00	0:30:00	89:59:57	89:59:57
Plan_ref_A1002k	Angle Y	0:00:00	-0:30:00	0:30:00	90:00:00	90:00:00
Plan_ref_A1002k	Angle Z	0:00:00	-0:30:00	0:30:00	0:00:03	0:00:03
Plan_ref_A1002k	Distance	31.000in	-0.100	0.100	30.974	-0.026
Plan_ref_G1004emod1Angle X	0:00:00	-0:30:00	0:30:00	89:59:49	89:59:49	89:59:49
Plan_ref_G1004emod1Angle Y	0:00:00	-0:30:00	0:30:00	89:59:13	89:59:13	89:59:13
Plan_ref_G1004emod1Angle Z	0:00:00	-0:30:00	0:30:00	179:59:14	179:59:14	179:59:14
Plan_ref_G1004emod1Distance	13.972in	-0.100	0.100	14.011	0.039	0.039
Plan_ref_G1004emod2Angle X	0:00:00	-0:30:00	0:30:00	90:00:01	90:00:01	90:00:01
Plan_ref_G1004emod2Angle Y	0:00:00	-0:30:00	0:30:00	90:00:01	90:00:01	90:00:01
Plan_ref_G1004emod2Angle Z	0:00:00	-0:30:00	0:30:00	179:58:59	179:58:59	179:58:59
Plan_ref_G1004emod2Distance	10.002in	-0.100	0.100	10.063	0.061	0.061
M1_D1_A1_C1002e	Position X-125.883	-0.100	0.100	-125.811	0.072	0.072
M1_D1_A1_C1002e	Position Y-126.371	-0.100	0.100	-126.412	-0.041	-0.041
M1_D1_A1_C1002e	Position Z12.727	-0.100	0.100	12.961	0.234	0.234
M1_D1_A2_C1003e	Position X-62.796	-0.100	0.100	-62.727	0.069	0.069
M1_D1_A2_C1003e	Position Y-123.104	-0.100	0.100	-123.074	0.030	0.030
M1_D1_A2_C1003e	Position Z12.727	-0.100	0.100	12.948	0.221	0.221
M1_D1_A3_C1004e	Position X-63.433	-0.100	0.100	-63.401	0.032	0.032
M1_D1_A3_C1004e	Position Y-93.361	-0.100	0.100	-93.331	0.030	0.030
M1_D1_A3_C1004e	Position Z12.727	-0.100	0.100	12.873	0.146	0.146
M1_D1_A4_C1005e	Position X-64.071	-0.100	0.100	-64.074	-0.003	-0.003
M1_D1_A4_C1005e	Position Y-61.617	-0.100	0.100	-61.587	0.030	0.030
M1_D1_A4_C1005e	Position Z12.727	-0.100	0.100	12.881	0.154	0.154
M1_D1_A5_C1006e	Position X-127.158	-0.100	0.100	-127.159	-0.001	-0.001
M1_D1_A5_C1006e	Position Y-62.884	-0.100	0.100	-62.925	0.041	0.041
M1_D1_A5_C1006e	Position Z12.727	-0.100	0.100	12.889	0.162	0.162
M1_D1_A6_C1007e	Position X-126.521	-0.100	0.100	-126.485	0.036	0.036
M1_D1_A6_C1007e	Position Y-94.627	-0.100	0.100	-94.669	-0.042	-0.042
M1_D1_A6_C1007e	Position Z12.727	-0.100	0.100	12.886	0.159	0.159



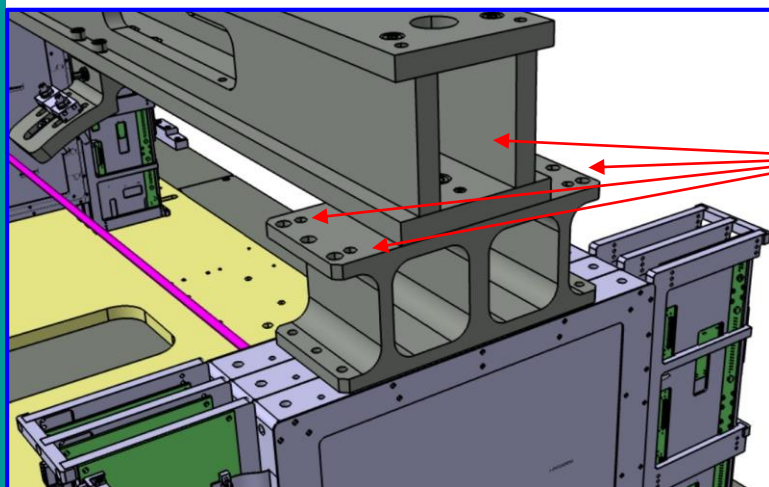
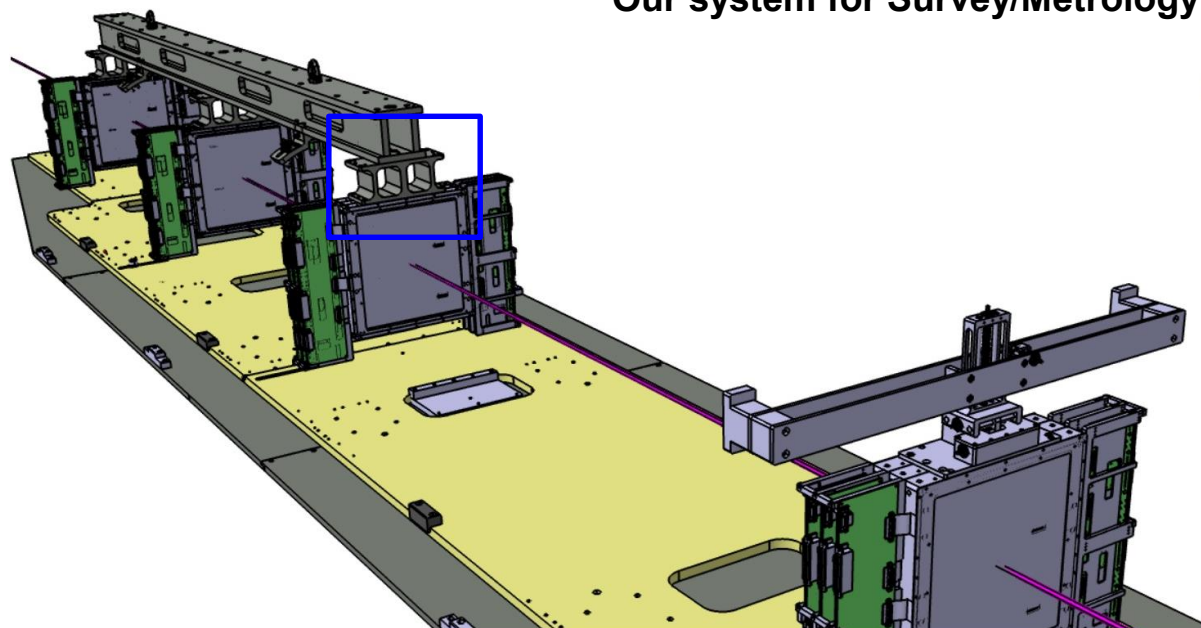
Metrology#2: IFT Mechanics

Tol.PosNoTol.ElemNaTol.ElemNTol.TolNameTol.NomiTol.LowerTolTol.UpperTolTol.Tol.Deviation						
Cyl_8H7_FrontLeft	Diamètre	8.000	0.000	0.012	8.028	0.028
Cyl_8H7_FrontLeft	Position	Y95.000	-0.050	0.050	95.031	0.031
Cyl_8H7_FrontLeft	Position	Z5.500	-0.050	0.050	5.482	-0.018
Cyl_8H7_BackLeft	Diamètre	8.000	0.000	0.012	8.019	0.019
Cyl_8H7_BackLeft	Position	Y70.000	-0.050	0.050	70.031	0.031
Cyl_8H7_BackLeft	Position	Z88.500	-0.050	0.050	88.494	-0.006
Cyl_8H7_BackRight	Diamètre	8.000	0.000	0.012	8.025	0.025
Cyl_8H7_BackRight	Position	Y70.000	-0.050	0.050	69.973	-0.027
Cyl_8H7_BackRight	Position	Z88.500	-0.050	0.050	88.473	-0.027
Cyl_8H7_FrontRight	Diamètre	8.000	0.000	0.012	8.022	0.022
Cyl_8H7_FrontRight	Position	Y95.000	-0.050	0.050	94.975	-0.025
Cyl_8H7_FrontRight	Position	Z5.500	-0.050	0.050	5.463	-0.037
Plan_Sup/AdjSystem	Planéité	0.000	0.000	0.050	0.067	0.067
Plan_Sup/AdjSystem	Position	X165.500	-0.100	0.100	165.696	0.196
Plan_Sup/AdjSystem	Perpendicu	0.000,	0.000	0.020	0.077	0.077



IFT Survey & Metrology

Our system for Survey/Metrology...



Metrology was done wrt I/F points

IFT Survey & Metrology

- File from survey (Camille), misalignments in meter / **nominal CAD positions**
- Measured points on Backbone (in blue, Station I/F's) and IFT (in red)
- Transfer to the station itself (metrology @UNIGE)

32											
33	Socket	FASER.BACKBONE.40.A	0,000398	0,010021	-0,0119	804600	477,77566	0,000398	0,000021	-0,011777	Measured Point
34	Socket	FASER.BACKBONE.40.E	0,000364	0,010038	-0,012	804601	477,77566	0,000364	0,000039	-0,011877	Measured Point
35	Socket	FASER.BACKBONE.40.F	0,000263	0,010064	-0,012209	804591	477,77566	0,000263	0,000065	-0,012086	Measured Point
36	Socket	FASER.BACKBONE.40.H	0,000223	0,009927	-0,011849	804588	477,77566	0,000223	-0,000072	-0,011726	Measured Point
37	Socket	FASER.BACKBONE.40.G	0,000235	0,010104	-0,012191	804590	477,83566	0,000235	0,000105	-0,012068	Measured Point
38	Socket	FASER.BACKBONE.40.I	0,000209	0,009984	-0,011841	804589	477,83566	0,000209	-0,000015	-0,011718	Measured Point
39	Socket	FASER.BACKBONE.40.J	0,000131	0,009989	-0,011925	804595	478,96566	0,000131	-0,00001	-0,011802	Measured Point
40	Socket	FASER.BACKBONE.40.L	0,000159	0,00987	-0,011815	804592	478,96566	0,000159	-0,000129	-0,011692	Measured Point
41	Socket	FASER.BACKBONE.40.M	0,000198	0,009925	-0,011873	804603	478,97566	0,000198	-0,000074	-0,01175	Measured Point
42	Socket	FASER.BACKBONE.40.T	0,000205	0,009896	-0,011833	804602	478,97566	0,000205	-0,000103	-0,01171	Measured Point
43	Socket	FASER.BACKBONE.40.C	0,000186	0,009901	-0,011851	804605	479,01566	0,000187	-0,000099	-0,011728	Measured Point
44	Socket	FASER.BACKBONE.40.D	0,000198	0,009961	-0,011782	804604	479,01566	0,000198	-0,000038	-0,011659	Measured Point
45	Socket	FASER.BACKBONE.40.K	0,000097	0,009963	-0,011937	804594	479,02566	0,000098	-0,000036	-0,011814	Measured Point
46	Socket	FASER.BACKBONE.40.N	0,000131	0,009856	-0,011818	804593	479,02566	0,000132	-0,000143	-0,011695	Measured Point
47	Socket	FASER.BACKBONE.40.O	-0,000141	0,009857	-0,011702	804599	480,15566	-0,000141	-0,000143	-0,011579	Measured Point
48	Socket	FASER.BACKBONE.40.Q	-0,000144	0,009734	-0,012042	804596	480,15566	-0,000144	-0,000265	-0,011919	Measured Point
49	Socket	FASER.BACKBONE.40.B	-0,000299	0,009731	-0,011795	804606	480,21566	-0,000299	-0,000268	-0,011672	Measured Point
50	Socket	FASER.BACKBONE.40.P	-0,00016	0,009825	-0,011694	804598	480,21566	-0,00016	-0,000174	-0,011571	Measured Point
51	Socket	FASER.BACKBONE.40.R	-0,000162	0,009724	-0,012034	804597	480,21566	-0,000162	-0,000275	-0,011911	Measured Point
52	Socket	FASER.BACKBONE.40.S	-0,000307	0,009792	-0,011735	804607	480,21566	-0,000307	-0,000208	-0,011612	Measured Point
53	Beam Point	FASER.BACKBONE.40.E	0,000334	0,009915	-0,011998	804391	477,75866	0,000334	-0,000085	-0,011875	
54	Beam Point	FASER.X1FDDT.20.E	0,000334	0,009915	-0,011998	801449	477,75866	0,000334	-0,000085	-0,011875	
55	Beam Point	FASER.X1FDDT.20.S	0,000308	0,009915	-0,011988	801450	477,85266	0,000308	-0,000085	-0,011865	
56	Beam Point	FASER.X1FDDT.40.E	0,000006	0,009915	-0,01187	801453	478,94866	0,000006	-0,000085	-0,011747	
57	Beam Point	FASER.X1FDDT.40.S	-0,00002	0,009915	-0,01186	801454	479,04266	-0,00002	-0,000085	-0,011737	
58	Beam Point	FASER.X1FDDT.60.E	-0,000321	0,009915	-0,011742	801457	480,13866	-0,000321	-0,000085	-0,011619	
59	Beam Point	FASER.BACKBONE.40.S	-0,000347	0,009915	-0,011732	804393	480,23266	-0,000347	-0,000085	-0,011609	
60	Beam Point	FASER.X1FDDT.60.S	-0,000347	0,009915	-0,011732	801458	480,23266	-0,000347	-0,000085	-0,011609	
61											

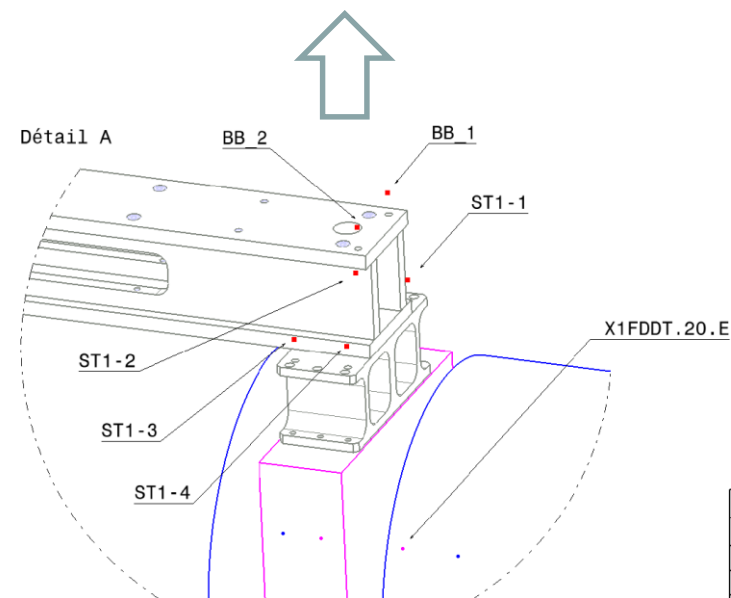
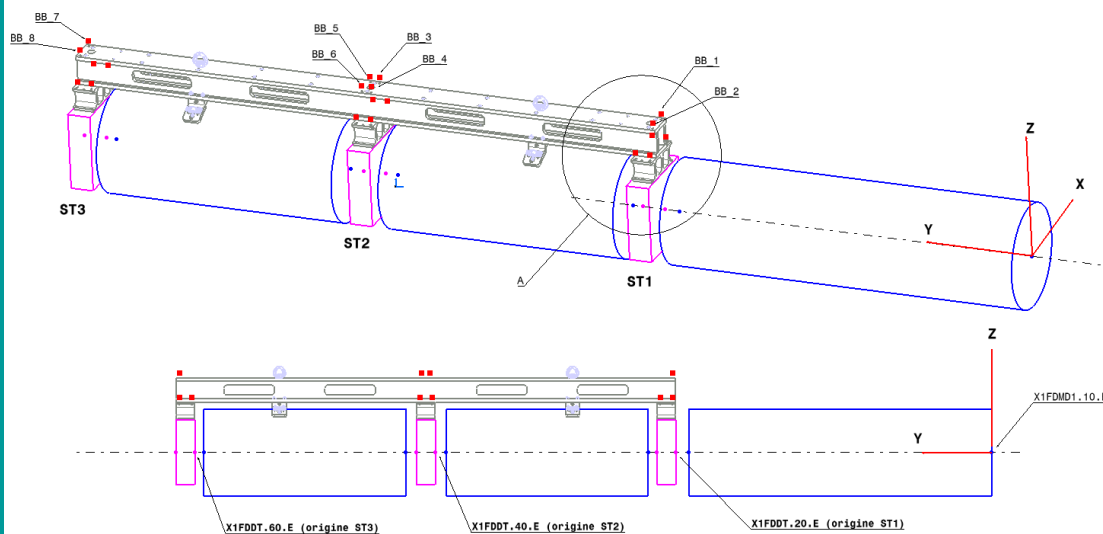
TRK_BackBone SURVEY



- Several targets onto the backBone (2 different areas)
- BackBone (BB#) & Station (ST#) related areas

Label correspondence

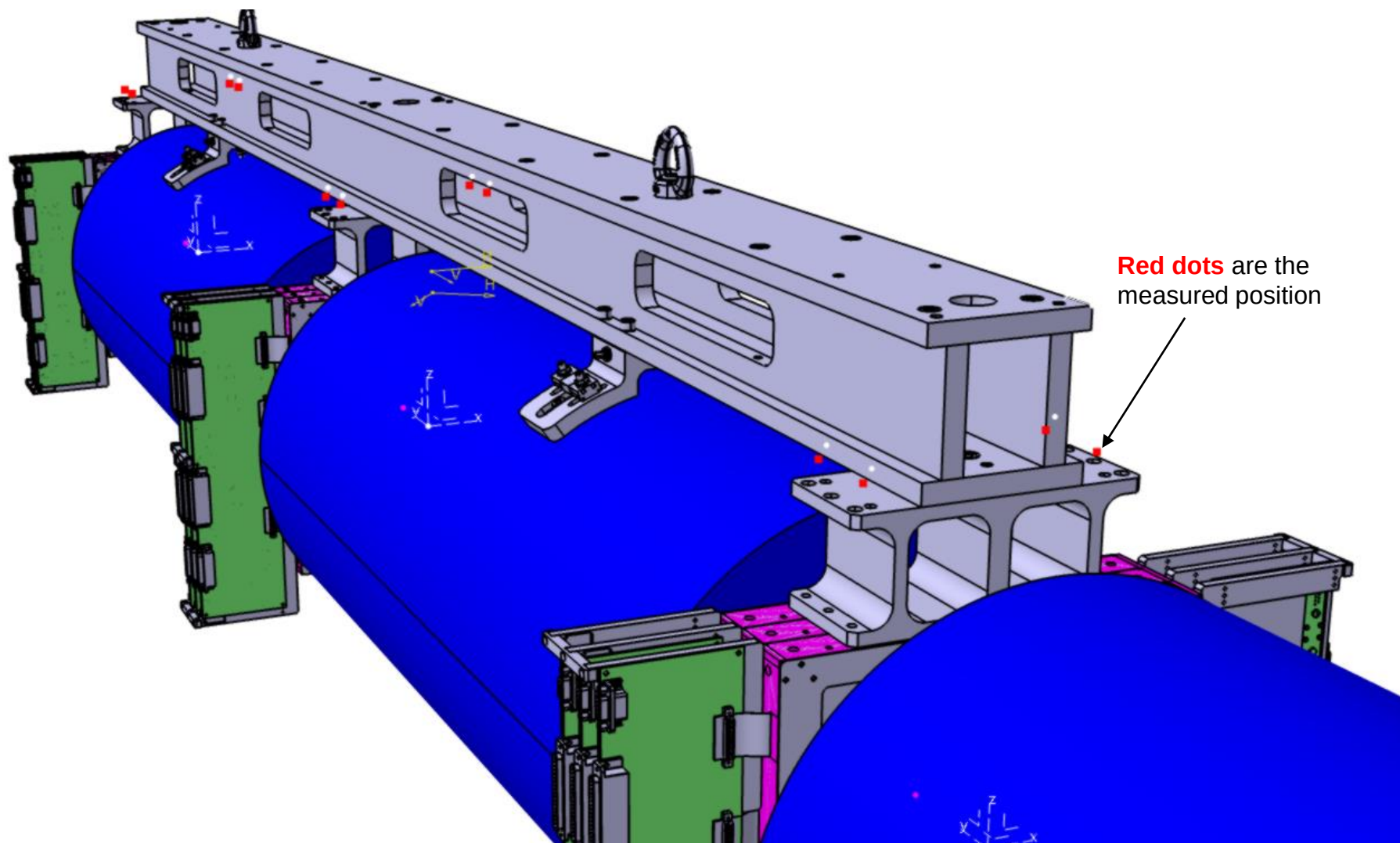
ST1-1	backbone.40.H
ST1-2	backbone.40.I
ST1-3	backbone.40.G
ST1-4	backbone.40.F
ST2-1	backbone.40.L
ST2-2	backbone.40.N
ST2-3	backbone.40.K
ST2-4	backbone.40.J
ST3-1	backbone.40.Q
ST3-2	backbone.40.R
ST3-3	backbone.40.P
ST3-4	backbone.40.O
BB_1	backbone.40.A
BB_2	backbone.40.E
BB_3	backbone.40.T
BB_4	backbone.40.M
BB_5	backbone.40.D
BB_6	backbone.40.C
BB_7	backbone.40.B
BB_8	backbone.40.S





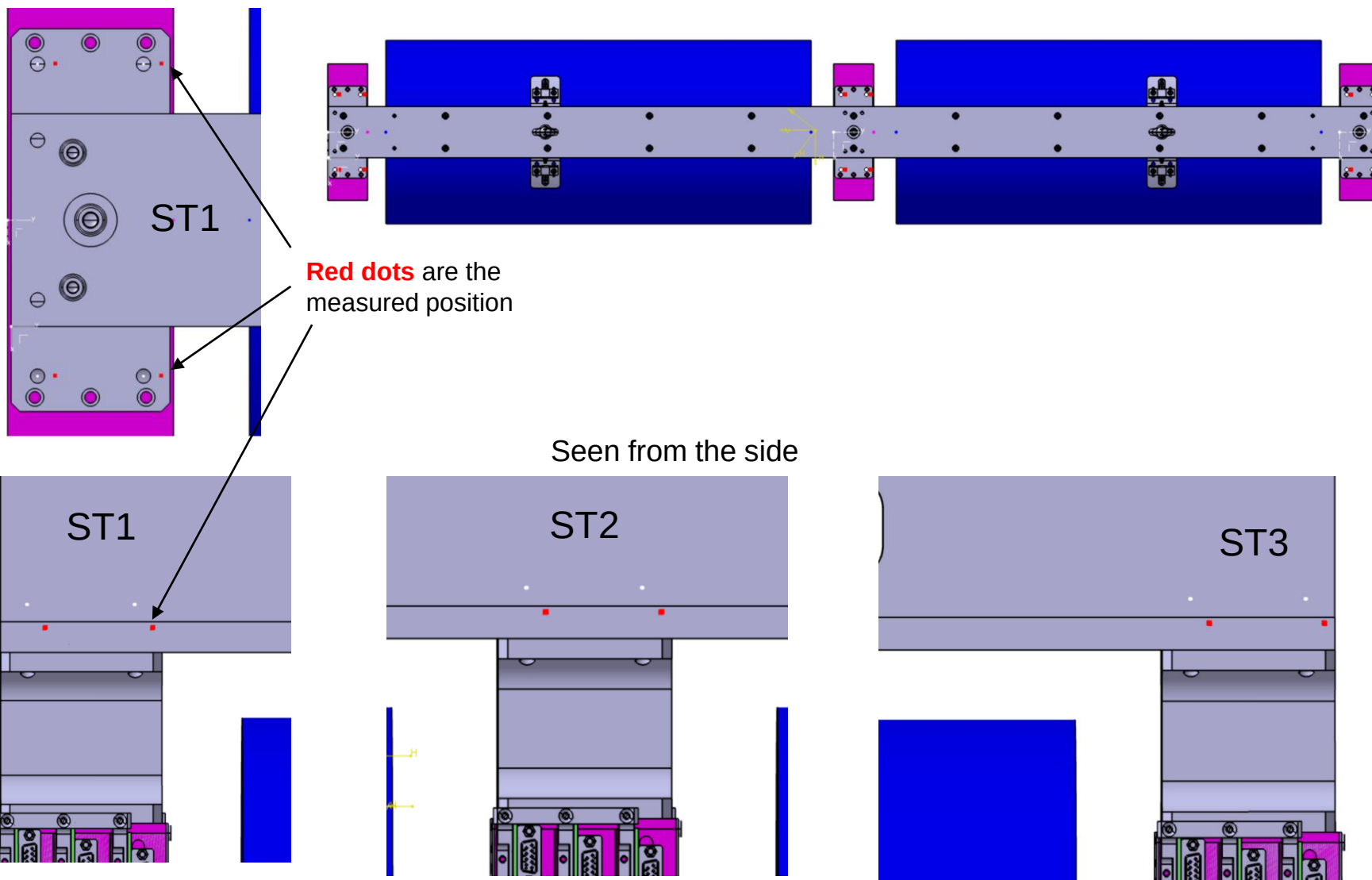
TRK_BackBone SURVEY

Preliminary analysis on the 3 stations (ST1, ST2 and ST3)



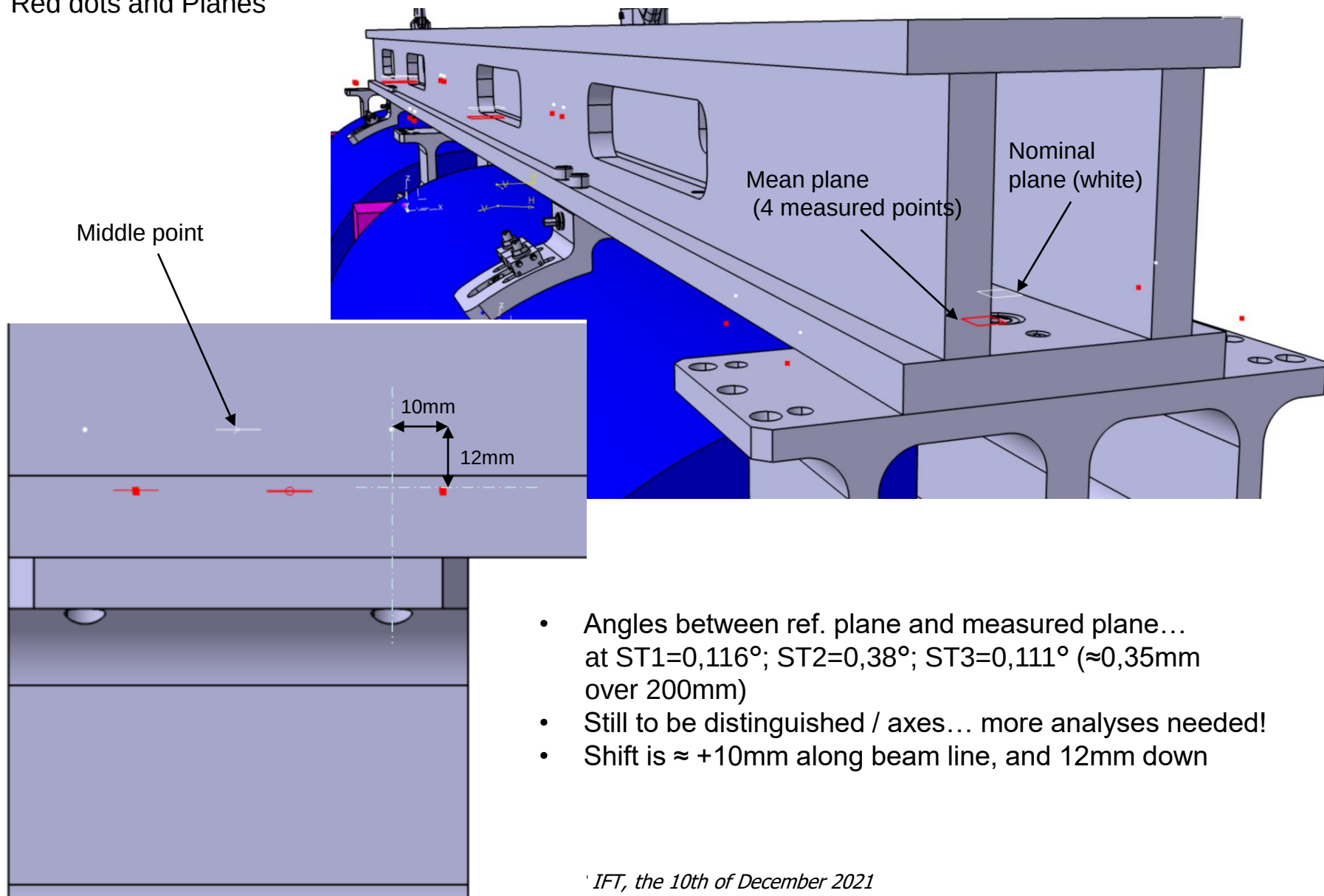


TRK_BackBone SURVEY



TRK_BackBone SURVEY

Red dots and Planes



- Angles between ref. plane and measured plane...
at ST1=0,116°; ST2=0,38°; ST3=0,111° ($\approx 0,35\text{mm}$ over 200mm)
- Still to be distinguished / axes... more analyses needed!
- Shift is $\approx +10\text{mm}$ along beam line, and 12mm down



CONCLUSIONS

NEXT STEPS:

- Combine all the files in something “readable”
- “Connect” the measured points (survey) with metrology (best fit?)
- Give all the sensor position / each others (OVER 3+1 STATIONS)